



• SOLAR ENERGY • LESSON 10

Solar Load Calculation

Load Schedule | Peak Sun Hours | System Sizing | Worked Example

What You Will Learn

- How to build a complete load schedule — identifying every load, watts, and daily hours
- Peak Sun Hours (PSH): what they are and how to find them for any location
- System sizing formulas for panels, batteries, MPPT controller, and inverter
- Full worked example: off-grid solar for a health clinic in Tigray, Ethiopia

Panel Size Formula

$$\text{Panel kWp} = \text{Daily Load (Wh)} \div (\text{PSH} \times \eta_{\text{system}})$$

η_{system} = overall system efficiency (typically 0.75 – 0.85)

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SOLAR-010: Solar Load Calculation

Every solar photovoltaic system starts with a question: how much energy does the load actually consume per day? Without an accurate answer, every downstream decision — panel size, battery capacity, inverter rating, cable sizing — is based on guesswork. An undersized system fails to power the load; an oversized system wastes money that could have been spent on better components or more loads. The solar load calculation is the foundation on which every professional solar design is built.

Learning Objectives

- Build a complete load schedule listing every electrical load with its rated power and daily usage
- Calculate total daily energy consumption in watt-hours (Wh/day) with appropriate safety margins
- Understand Peak Sun Hours (PSH) and determine PSH values for any location
- Apply system sizing formulas to determine required panel kWp, battery kWh, MPPT rating, and inverter size
- Complete a full load calculation for a real off-grid application

1. Why Load Calculation Comes First

Many first-time solar designers make the mistake of starting with panel size — "I want a 5 kWp system" — without knowing what the load actually needs. This leads to a fundamental mismatch between generation and consumption. The correct sequence always starts with the load.

Design Step	Input Needed	Output Produced
Step 1: Load schedule	List of all electrical loads, their wattage, and daily use hours	Daily energy consumption (Wh/day)
Step 2: Apply margins	Daily energy, system efficiency, safety factor	Design energy target (Wh/day after losses)
Step 3: Panel sizing	Design energy target, Peak Sun Hours (PSH) for location	Required panel capacity (kWp)
Step 4: Battery sizing	Design energy, days of autonomy required, depth of discharge (DoD)	Battery bank capacity (kWh or Ah at system voltage)
Step 5: MPPT sizing	Panel array Isc and Voc, battery voltage	MPPT controller current rating (A)
Step 6: Inverter sizing	Total simultaneous peak load (W)	Inverter rated power (kW) and surge rating

2. Building the Load Schedule — Step by Step

The load schedule is a systematic table listing every electrical device that will be powered by the solar system. For each load, you need: **rated power (W)**, **quantity**, **hours of use per day**, and **days of use per week**. From these four values, you can calculate the daily energy contribution of each load.

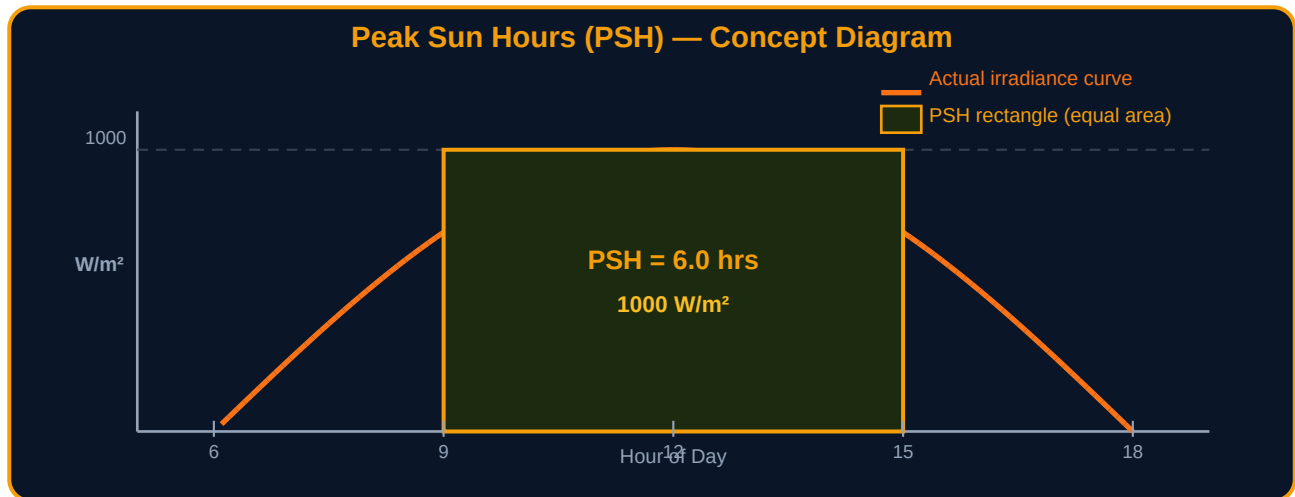
Load Schedule Formula

$$\text{Daily Energy (Wh/day)} = \text{Power (W)} \times \text{Quantity} \times \text{Hours/Day} \times (\text{Days/Week} \div 7)$$

Important: Use the actual *operating power*, not the nameplate rating, for loads that do not run at full rated power continuously. A refrigerator rated 150 W does not run 24 hours — its compressor cycles on/off. Use the *daily energy from the manufacturer datasheet* (kWh/24h) or multiply rated power by the *duty cycle* (fraction of time running).

3. Peak Sun Hours (PSH)

Peak Sun Hours (PSH) is a key parameter that relates your location's available solar resource to the performance of your solar panels. One PSH = one hour of sunlight at a radiation intensity of 1000 W/m^2 (which is also the Standard Test Condition for solar panel ratings). A location with $\text{PSH} = 5$ means the total daily solar radiation is equivalent to 5 hours at 1000 W/m^2 .



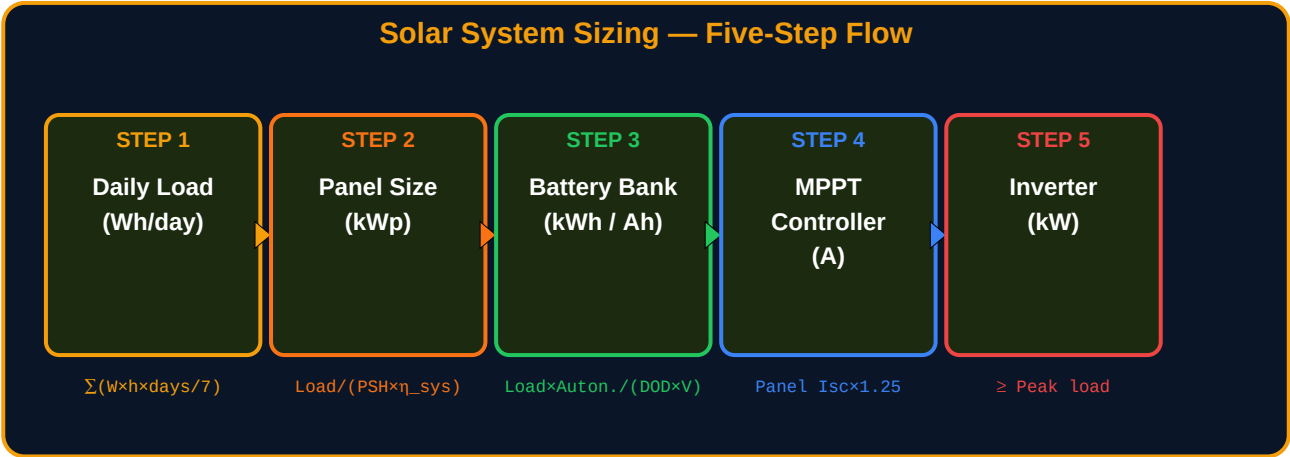
PSH Values for Key African and Middle East Locations

Location	Country	PSH (Annual Average)	Best Month	Design PSH (use worst month)
Mekelle / Tigray	Ethiopia	6.0 — 6.5 h/day	November (7.2)	5.5 (July — rainy season)
Addis Ababa	Ethiopia	5.5 — 6.0 h/day	November (6.8)	5.0 (July)
Nairobi	Kenya	5.5 — 6.2 h/day	January (6.5)	4.8 (June)
Lagos	Nigeria	4.5 — 5.2 h/day	January (5.8)	3.8 (August)
Cairo	Egypt	5.8 — 6.5 h/day	June (8.2)	5.2 (December)
Dubai	UAE	5.5 — 6.8 h/day	June (7.5)	5.0 (December)
Johannesburg	South Africa	5.5 — 6.5 h/day	December (7.0)	4.8 (June)
Kigali	Rwanda	4.8 — 5.5 h/day	June (5.8)	4.2 (April — long rains)

Always design using the **worst-month PSH** (lowest value) — this ensures the system meets load requirements even in the least favourable month. If you design for average PSH, the system will under-perform for several months of the

year.

4. System Sizing Overview



Component	Sizing Formula	Key Parameters	Typical Value
Solar panels (kWp)	$\text{kWp} = E_{\text{design}} \div (\text{PSH} \times \eta_{\text{system}})$	$\eta_{\text{system}} = 0.75\text{--}0.80$ (includes wiring, MPPT, inverter, temperature losses)	1 kWp per ~900 Wh/day in 6h PSH location
Battery bank (kWh)	$\text{kWh} = (E_{\text{daily}} \times \text{Autonomy}) \div \text{DoD}$	Autonomy = days of storage (1–3); DoD = 0.8 LiFePO4, 0.5 Lead-Acid	LiFePO4: 1 kWh stores ~0.8 kWh usable
Battery Ah (at system V)	$\text{Ah} = \text{kWh} \times 1000 \div V_{\text{system}}$	$V_{\text{system}} = 12\text{V}, 24\text{V}, \text{ or } 48\text{V}$. Use 48V for systems above 2 kWp	100 Ah at 48V = 4.8 kWh capacity
MPPT controller (A)	$I_{\text{mppt}} = (\text{Panel I}_{\text{sc}} \times 1.25)$ at battery voltage	I_{sc} = short-circuit current; 1.25 = safety factor per NEC/IEC	Select next standard size above calculated
Inverter (kW)	$\text{kW} \geq \text{Total simultaneous peak load}$	Add all loads that may run at the same time; select inverter with 2× surge rating for motor start	Min: $1.25 \times \text{peak simultaneous load}$

5. Worked Example — Off-Grid Health Clinic, Tigray, Ethiopia

Scenario: A rural health clinic in Hawzen district, Tigray, requires an off-grid solar system. The clinic operates 7 days/week. Design month: July (worst month PSH = 5.5 h/day). System voltage: 48V DC. Battery: LiFePO4 (DoD = 0.8, 2 days autonomy). System efficiency $\eta = 0.78$.

Step 1 — Load Schedule

Load	Power (W)	Qty	Hrs/Day	Days/Wk	Daily Energy (Wh/day)
LED ceiling lights (ward)	12	8	10	7	$8 \times 12 \times 10 \times 7/7 = 960$
LED ceiling lights (OPD/offices)	12	6	8	7	$6 \times 12 \times 8 = 576$
Medical refrigerator (vaccine)	60	1	24	7	$60 \times 24 \times (\text{duty cycle } 0.5) = 720$
Desktop computer + monitor	130	2	8	6	$2 \times 130 \times 8 \times 6/7 = 1,783$
Phone/tablet charging station	60	1	6	7	$60 \times 6 = 360$
Water pump (solar-compatible)	370	1	2	7	$370 \times 2 = 740$
Ultrasound machine	200	1	3	5	$200 \times 3 \times 5/7 = 429$
Miscellaneous / spare	—	—	—	—	300 (10% buffer)
TOTAL RAW DAILY ENERGY					5,868 Wh/day

Step 2 — Apply Safety Margin

Calculation	Formula	Result
Raw daily energy	Sum of all loads	5,868 Wh/day
System losses / inefficiency	$\div \eta_{\text{system}} = 0.78$	$5,868 \div 0.78 = 7,523 \text{ Wh/day}$
Additional safety margin (10%)	$\times 1.10$	$7,523 \times 1.10 = 8,275 \text{ Wh/day}$
Design energy target (round up)	Use for all further calculations	8,300 Wh/day $\approx$ 8.3 kWh/day

Step 3 — Panel Sizing

$$\text{Panel kWp} = E_{\text{design}} \div \text{PSH} = 8,300 \div 5.5 = 1,509 \text{ W} \approx 1.5 \text{ kWp}$$

Select standard panels: **6 × 280 Wp panels = 1,680 Wp (1.68 kWp)** — next standard array size above 1,509 Wp. This 12% oversize provides a buffer for panel degradation (0.5%/year) and occasional heavy clouding.

Step 4 — Battery Bank Sizing

$$\text{Battery kWh} = (E_{\text{daily}} \times \text{Autonomy}) \div \text{DoD} = (8.3 \times 2) \div 0.8 = 20.75 \text{ kWh}$$

$$\text{Battery Ah at 48V} = 20,750 \div 48 = 432 \text{ Ah}$$

Select: **8 × 100 Ah LiFePO4 batteries at 12V, wired 4S2P to give 48V, 200 Ah = 9.6 kWh usable.** Or: **4 × 200 Ah at 12V in 4S = 48V, 200 Ah = 9.6 kWh.** LiFePO4 batteries at 80% DoD: usable = $9.6 \times 0.8 = 7.68 \text{ kWh}$ — provides 22 hours at design load. Adjust number of batteries based on available budget and space.

Step 5 — MPPT Controller Sizing

Parameter	Calculation	Value
Panel array configuration	6 panels $\times$ 280 Wp $\rightarrow$ arrange as 2S3P (2 strings of 3 panels)	2 strings in parallel, 3 panels per string
String voltage (Voc)	3 $\times$ panel Voc (typically 40V) = 120V per string	Array Voc = 120V
Array Isc	2 strings $\times$ panel Isc (typically 9.5A) per string	Array Isc = 19.0A
MPPT input current (with 1.25 safety)	19.0 $\times$ 1.25 = 23.75A minimum	Select 30A MPPT controller
MPPT voltage range check	Input Voc (120V) must be within MPPT voltage window	Verify 120V < MPPT max input voltage (e.g. 150V or 200V)

Step 6 — Inverter Selection

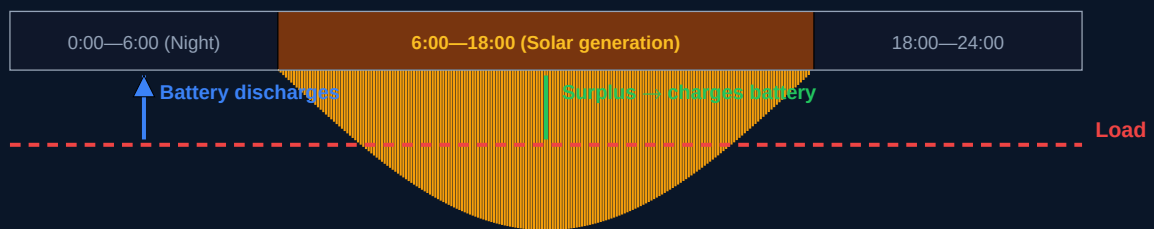
Loads potentially running simultaneously	Power (W)
Ward lights + OPD lights (50% coincidence)	(960+576) $\times$ 0.5 / 10hrs + on at once = 576W lighting max
Medical refrigerator (always on)	60 W
2 computers (both running)	260 W
Water pump start surge (motor $\times$ 3 for start)	370 $\times$ 3 = 1,110 W peak (momentary)
Total simultaneous peak (excluding pump surge)	$\approx$ 900 W continuous
Inverter selection	1,500 W continuous with 3,000 W surge (to handle pump start)

System Summary

Component	Specification Selected	Quantity
Solar panels	280 Wp monocrystalline, 40V Voc, 9.5A Isc	6 panels (1.68 kWp)
Battery bank	200 Ah LiFePO4 at 12V, wired 4S = 48V, 200 Ah	4 batteries (9.6 kWh)

MPPT charge controller	48V system, 30A, 150V max input, MPPT algorithm	1 unit
Inverter/charger	48V DC input, 1,500W continuous, 3,000W surge, pure sine wave	1 unit
DC distribution	48V DC busbar, 63A fuse, cable sizing per IEC 62548	Per schematic
Earthing	System earth rod + panel frame bonding per IEC 62548 / IEC 60364	Per site survey

Energy Balance — Generation vs Consumption



System is balanced when: Panel generation $\geq$ Daily load $\div$ System efficiency

Battery stores surplus midday energy for use at night and cloudy days

6. Practice Exercises

#	Exercise	Data / Instructions
1	Load schedule for a house	Build a load schedule for a 3-bedroom house: 10 LED lights (9W, 6h/day), TV (80W, 5h/day), fridge (150W, 24h, duty cycle 0.4), fan (60W, 8h/day), phone chargers (40W, 3h/day). Calculate total daily energy in Wh/day.
2	Panel sizing	Daily design energy = 4,500 Wh/day. PSH = 5.5h. System efficiency = 0.78. Calculate required panel kWp. If 300W panels are available, how many panels are needed?
3	Battery sizing	Daily energy = 6,000 Wh/day. 2 days autonomy. LiFePO4 batteries (DoD = 0.8). System voltage = 48V. Calculate required battery kWh and Ah. If 200Ah 12V batteries are used, how many in what configuration?
4	Worst-month PSH effect	A system is sized for Addis Ababa average PSH = 5.8h. Load = 5,000 Wh/day. The system works fine 10 months of the year but underperforms in July (PSH = 4.5h). Calculate: (a) panel size for average PSH, (b) panel size for worst-month PSH. What is the % difference? Why does designing for worst-month matter?
5	MPPT and inverter selection	Panel array: 4 × 360W panels, Voc=45V, Isc=9.8A, wired 2S2P. System: 24V. Loads: 2 fans (60W each), lights (120W), laptop (65W). Size the MPPT controller and inverter for this system.

Lesson Summary

Concept	Key Rule
Load schedule first	Always start with the load. Never size panels before knowing daily energy consumption.
Daily energy (Wh/day)	Sum of (Watts × Hours/day × Quantity) for every load, 7 days normalised.
Design energy	Raw energy ÷ system efficiency (0.75–0.80) × safety margin (1.10). Always use this, not raw energy, for sizing.
Worst-month PSH	Always design using the lowest-PSH month. Average PSH leads to under-performance for several months per year.
Panel kWp formula	$\text{kWp} = \text{Design energy} \div (\text{PSH} \times \text{system efficiency})$. Convert to number of panels: round UP to next whole panel.
Battery kWh formula	$\text{kWh} = (\text{Daily energy} \times \text{Autonomy days}) \div \text{DoD}$. Autonomy = 1 day for grid-hybrid, 2–3 days for pure off-grid critical loads.
MPPT sizing	Current = Array Isc × 1.25. Voltage = check array Voc against MPPT max input voltage. Always verify both.
Inverter sizing	Must handle all simultaneous continuous loads PLUS motor start surges (typically 3× rated power momentary). Select 1.25× peak continuous load minimum.

What Is Next — SOLAR-011

SOLAR-011: Battery Sizing — Deep dive into battery selection and sizing. Lead-acid vs LiFePO4 vs lithium NMC chemistry comparison. C-rate, depth of discharge, cycle life, and temperature effects. Series and parallel battery configurations for 12V, 24V, and 48V systems. Battery management systems (BMS). Complete battery bank sizing methodology with worked examples.

Resource	Details
SOLAR PDF Series	SOLAR-001 through SOLAR-030 — ayetechub.com/pdfs.html
Solar PV Sizing Calculator	Free interactive tool — ayetechub.com/downloads/solar-calculator.html
Community	Telegram: t.me/ayetechub